

Protocol-aware wind-power event measurement

China
4 archives · 298 turbines

Europe
4 archives · 42 turbines

Physical time · quality flags · training-only scaling

1 Detection



Ramps · V / inverted V

2 Correspondence



One-to-one + overlap graph

3 Representation



Raw · PCA · angular + sign

4 Evaluation and decision

Structural survival

ARI · NMI · coverage
Split / merge composition

Physical tracking

ERA5 · independent LiDAR
Direction · calibrated
probability

Forecast-to-cost

Issued forecast → settlement
Debit · credit · storage

Independent references
Blind ratings · weather · LiDAR · prices